

Adaptive Confidence-Weighted Bearing Formation Control with Quaternion-Based Attitude Synchronization for UAV Swarms

~ Introduction to Drones – 22AIE448

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Motivation

The growing adoption of UAV swarms in disaster response, surveillance, precision agriculture, and infrastructure inspection demands reliable and robust formation control. However, real-world operations involve sensing uncertainties, communication degradation, and dynamic environments, creating a need for adaptive control strategies that can maintain stable and coordinated formations under practical operating conditions.

Research Gap

- **Gap 1: Equal Neighbor Weighting** – Existing bearing controllers assign equal importance to all neighboring UAVs, ignoring variations in sensing quality and communication reliability.
- **Gap 2: No Bearing Confidence Estimation** – Current methods assume bearing measurements are perfectly reliable despite sensor noise and uncertainties.
- **Gap 3: Ignored Rigidity Importance** – Conventional controllers treat all graph edges equally, overlooking their structural significance.

Problem Statement

Existing bearing-based UAV formation controllers **equally weight** neighboring information, **ignore bearing measurement uncertainty**, and **overlook edge rigidity importance**, resulting in reduced robustness, slower convergence, and degraded formation stability under real-world conditions.

Bearing - Relative Direction (unit vector) and Orientation (angle)

Objectives

- To develop a covariance-based Bearing Confidence Estimator that quantifies the reliability of relative bearing measurements under sensing uncertainties.
- To design an adaptive confidence- and rigidity-weighted bearing formation controller that dynamically adjusts each neighbor's influence based on measurement confidence and structural importance.
- To evaluate the proposed controller through simulation by comparing its formation convergence, stability, and robustness against the conventional bearing formation controller under noisy operating conditions.

Literature Review

Year	Paper	Main Contribution	Methodology Used	Limitations
2022	<u>Bearing-Based Distributed Formation Control of UAV Swarm by Quaternion-Based Attitude Synchronization</u>	Integrated bearing-based formation control with quaternion attitude synchronization for stable 3D UAV swarm coordination.	Bearing controller, quaternion attitude control, distributed control, Lyapunov stability analysis.	Equal neighbor weighting, no bearing confidence estimation, and no rigidity-aware adaptive control.
2026	Bearing-Based Multi-Agent Formation Control: A Survey and Taxonomy	Reviewed recent advances in bearing-based formation control and identified future research directions for robust distributed UAV coordination.	Systematic literature review and taxonomy of bearing-based control methods.	Highlights unresolved issues including measurement uncertainty, adaptive weighting, and robustness to sensing and communication degradation.

Year	Paper	Main Contribution	Methodology Used	Limitations
2012	Formation Shape Control Based on Bearing Rigidity	Established the theoretical foundation of bearing rigidity for preserving formation shape using bearing constraints instead of distances.	Bearing rigidity theory, graph theory, distributed formation control.	No UAV dynamics, no uncertainty modeling, all bearing constraints treated equally.
2016	A Rigidity-Based Decentralized Bearing Formation Controller for Groups of Quadrotor UAVs	Proposed a decentralized bearing-based formation controller for quadrotor UAVs using local visual measurements.	Bearing rigidity, distributed control, onboard vision, graph-based formation control.	Equal influence assigned to all neighboring UAVs; no confidence estimation or adaptive weighting.
2021	Distributed Bearing-Based Formation Control of unmanned aerial vehicle swarm via Global Orientation Estimation	Eliminated the need for a shared global coordinate frame by estimating orientation from local measurements.	Bearing estimation, distributed orientation estimation, graph theory, consensus control.	Assumes accurate bearing measurements and ignores sensing uncertainty and edge importance.

Action Plan

Phase	Activities	Deliverables
Formation Modeling & Bearing Analysis	Develop the formation graph using Python, NumPy, and NetworkX. Implement node/edge representation, bearing vector computation, projection matrix, and verify formation convergence using the standard bearing controller.	Functional graph-based bearing formation model with validated convergence.
Adaptive Confidence-Weighted Bearing Controller	Design the Bearing Confidence Estimator, Rigidity Contribution Analyzer, Adaptive Weight Generator, and implement the proposed confidence- and rigidity-weighted bearing controller. Compare its performance with the conventional controller.	Novel adaptive bearing controller with improved convergence and robustness under noisy conditions.
UAV Dynamics & Quaternion Integration	Implement quaternion-based attitude control and integrate it with UAV translational dynamics, including position, velocity, acceleration, angular velocity, and orientation. Connect the adaptive controller with the quadrotor dynamic model.	Complete UAV control architecture integrating adaptive bearing formation with quaternion attitude synchronization.
Simulation & Performance Evaluation	Integrate the controller with ROS2, PX4 SITL, and Gazebo (or AirSim/Webots). Simulate multiple UAVs, inject sensing and communication disturbances, and evaluate convergence time, bearing error, stability, and robustness against the base controller.	Validated multi-UAV simulation demonstrating the effectiveness of the proposed controller under realistic operating conditions.

Tech Stack & Concepts

- Programming Language - Python
- Scientific Computing - NumPy
- Graph and Formation Modelling - NetworkX
- Control Maths - Graph Theory / Quaternions
- Controller Development - Adaptive Bearing / Quaternion Attitude controller
- Simulation - Webots
- Drone - TBD

Proposed Methodology

- **Define the desired UAV formation** as a graph, where each UAV is represented as a node and the desired formation is specified using inter-agent bearing vectors.
- **Estimate relative bearing vectors** between neighboring UAVs using their relative positions and compute the corresponding projection matrices required for bearing-based formation control.
- **Develop a Bearing Confidence Estimator** that quantifies the reliability of each bearing measurement by estimating its covariance under sensing uncertainties or injected Gaussian noise.
- **Analyze graph rigidity contribution** by evaluating the structural importance of each communication edge and assigning a rigidity score based on its impact on formation integrity.

Proposed Methodology

- **Generate adaptive edge weights** by combining the bearing confidence and rigidity score to determine the influence of each neighboring UAV in the control law.
- **Implement the proposed confidence-weighted bearing controller**, where adaptive weights dynamically modify the conventional bearing control input to improve convergence and robustness.
- **Integrate the controller with quaternion-based attitude synchronization** to compute the required orientation and rotational motion while avoiding singularities associated with Euler angles.
- **Validate the complete controller** through multi-UAV simulations in a ROS2–PX4–Gazebo (or AirSim/Webots) environment and compare its performance with the conventional bearing controller using metrics such as formation convergence, bearing error, stability, and robustness under noisy conditions.

Expected Outcomes & Feasibility Study

- Develop a **novel adaptive confidence- and rigidity-weighted bearing formation controller** for distributed UAV swarms.
- Achieve **faster formation convergence** and **reduced oscillations** compared to the conventional bearing controller.
- Improve **robustness against sensor noise, communication degradation, and bearing uncertainties**.
- Successfully integrate **quaternion-based attitude synchronization** for smooth 3D UAV orientation control.
- Validate the proposed controller through **multi-UAV simulation** and benchmark it against the base controller.

Thank You